MILITARY SPECIFICATION

PROJECTILE, 20MM, ARMOR PIERCING-TRACER, M95, METAL PARTS

This specification is approved for use within the U.S Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, preparation for delivery of metal parts for Projectile, 20MM, Armor Piercing Tracer (APT), M95.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. The following specifications, and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-12560 - Armor Plate, Steel, Wrought, Homogenous (for use in Combat-Vehicle Ammunition Testing)

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament,Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305

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MIL-STD-105-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-651-	Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1168-	Lot Numbering of Ammunition
MIL-STD-1169-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS (SEE 6.7)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

Product and Packaging Drawings

7258816	- Blank, Rotating Band
7258861-5	- Carton, for Interplant Shipment of Projectiles, 20mm, Assemblies and Metal Parts Assemblies (200)
7259136-5	- Shipping Container for Inert Projectile, 20mm Assemblies and Metal Parts Assemblies (10,000)
12576577	- Projectile, 20mm, Armor Piercing Tracer, M95, M/P
12576578	- Body and Windshield Assy, Proj., 20mm, AP-T, M95
12576579	- Body and Rotating Band Assy, Proj., 20mm AP-T, M95
12576580	- Body, Shot, Proj., 20mm, Armor Piercing Tracer, M95
12576581	- Windshield, Proj., 20mm, AP-T, M95
7259190-1	- Cartridge, 20mm for M90 Type Projectile Metal Parts Test

Inspection Equipment Drawings

5055857	- Comb Ring and Snap Gage
7259353	- Go, Flat Plug Gage, Special
7259467	- Comparator, Optical, Hand Type
7259536	- Not Go, Flat Plug Gage, Special
7259545	- Operating Instructions (Band Tester)
7259718	- Optical Gaging Chart
7275149	- Step Plug Gage
7286995	- Flush Pin Gage Assembly
7287005	- Flush Pin Gage
7558294	- Staging Fixture, Gage

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7558875 - Indicator Gage
7558876 - Acceptance Check
8655592 - Special Go Plain Ring Gage
8657201 - Optical Gaging Chart
11075094 - Gage, Snap, Go, Special
11075203 - Eddy Current Flaw Detector

PUBLICATIONS

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges.

U.S. ARMY TEST AND EVALUATION COMMAND

PUBLICATIONS

TECOM TOP 2-2-710 - Ballistic Test of Armor Materials

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 - Standard Method of Salt Spray (Fog) Testing
ASTM-E18 - Method of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General.

a Material. Materials shall be in accordance with the applicable drawings and specifications.

b. Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval (6.4).

3.2 Components. All parts and assemblies shall comply with all requirements specified on Drawing (Dwg.) 12576577, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article . When specified in the contract or purchase order, a sample shall be subjected to a first article inspection in accordance with the technical provisions herein (see 4.3 and 6.5).

3.4 Banding. The rotating band blank and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used to clean the rotating band seat after knurling. The rotating band blank shall fill the band seat.

3.5 Soundness. Metal dicontinuity in the projectile body as determined by eddy current inspection shall be less than that represented by production check standard (PCS) No 1.

3.6 Protective coating.

3.6.1 Phosphate coating. The pretreatment cleaning process shall be in accordance with the applicable requirements of Dwg. 12576577. The phosphate coating shall be in accordance with the requirements of TT-C-490 for appearance and minimum weight as applicable to the type of phosphate coating used. Test specimens (panels) shall be in accordance with the requirements of TT-C-490.

3.6.2 Paint coating. The exterior coating of the projectile shall be in accordance with applicable requirements of Dwg. 12576577. The exterior paint adhesion shall be in accordance with applicable requirements of TT-C-490.

3.6.3 Corrosion resistance. The exterior painted surface of the projectile shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

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3.7 Metal parts security. The projectile metal parts when assembled into test cartridges in accordance with Dwg. 7259190-1 shall not separate or break up in the test for metal parts security.

3.8 Hardness. The shot body shall be in accordance with the respective hardness requirements specified on Dwg. 12576580.

3.9 Windshield.

3.9.1 Cleaning before assembly. The projectile body and windshield shall be cleaned and free from foreign material at the time of attaching the windshield to the shot body.

3.9.2 Windshield attachment. The windshield shall be rigidly and firmly attached thereto in the manner prescribed by the applicable Dwg. and shall be concentric with the projectile body within the limits prescribed on the drawing. The assembly shall be free of cracks and tears.

3.9.3 Windshield torque. Windshield shall withstand a 1000 inch-pound torque without movement of the windshield relative to the shot body.

3.10 Penetration. The projectile shall exhibit a V50 protection ballistic limit BL (P) of 1960 + 25 feet per second against a 3/4 inch homogeneous armor plate conforming to MIL-A-12560 at 0° obliquity.

3.11 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Processing defects. The projectile shall be free of defects, such as dents, deep scratches, cuts, excessive surface roughness, metal flaws, burrs, sharp edges, and other forms of substandard workmanship.

3.11.2 Cleanliness. The projectile shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

a. Inspection and nonfiring tests:

Part Description	Drawing	Quantity	Method Para
Standard test panel		3	4.5.5
Rotating band	7258816	32	4.4.2.1
Windshield	12576581	125	4.4.2.2
Shot Body	12576580	125	4.4.2.3
Body & Rotating Band Assy	12576579	125	4.4.2.4
Body & Windshield Assy	12576578	125	4.4.2.5
Projectile (painted)	12576577	170	4.4.2.6

b. Firing tests:

Part Description	Drawings	Quantity	Requirement & Method Para
Projectile, Metal Parts	12576577	4000	3.7/4.5.8
Projectile, Metal Parts	12576577	--	3.10/4.5.11

c. The projectile Dwg. 12576577 shall be identified in accordance with MIL-STD-1168.

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4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with the applicable drawings and requirements of Table I, as specified herein, will result in sample disapproval.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70637(AR)

PARAGRAPH	TITLE: Projectile, 20MM, AP-T, M95, Metal Parts		SHEET 1 OF 2		DRAWING NUMBER: SEE BELOW
					NEXT HIGHER ASSEMBLY:
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotating Bands - Dwg. 7258816		Acc-0		
	Examination for defects	32	Rej-1	3.2	4.4.2.1
	Windshield - Dwg. 12576581		Acc-0		
	Examination for defects	125	Rej-1	3.2	4.4.2.2
	Shot Body - Dwg. 12576580		Acc-0		
	Examination for defects	125	Rej-1	3.2	4.4.2.3
	Hardness	13	Acc-0		
			Rej-1	3.8	4.5.9.1
	Body and Rotating Band Assy - Dwg. 12576579		Acc-0		
	Examination for defects	125	Rej-1	3.2	4.4.2.4
	Body and Windshield Assy - Dwg. 12576579		Acc-0		
	Examination for defects	125	Rej-1	3.2	4.4.2.5
	Soundness verification	125	Acc-0		
			Rej-1	3.5	4.5.4
	Automatic Inspection effectiveness			3.5	4.5.3
	Windshield torque		Acc-0		
		20	Rej-1	3.9	4.5.10

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection

MIL-P-70637(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile, 20MM, AP-T, M95, Metal Parts				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Phosphate Coating		Acc-0		
	Standard test panels (or specimens)	3	Rej-1	3.6.1	4.5.5
	Projectile (painted) - Dwg. 12576577		Acc-0		
	Examination for defects	125	Rej-1	3.2	4.4.2.6
	Band tightness	125	Acc-0		
			Rej-1	3.4	4.5.1
	Paint adhesion	13	Acc-0		
			Rej-1	3.6.2	4.5.6
	Corrosion resistance	32	Acc-0		
			Rej-1	3.6.3	4.5.7
	Firing Tests				
	Metal parts security	4000	Acc-0		
			Rej-1	3.7	4.5.8
	Penetration	1/	2/	3.10	4.5.11

NOTES: 1/ Quantity as required to determine V50.
2/ A V50 value equal to or more than that specified in 3.10 shall be cause for rejection of the first article quantity.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Major and minor defects. To determine product acceptability, major and minor defects as listed herein may be considered collectively on a class basis or defects may be considered individually. When a class basis is utilized the Acceptable Quality Level (AQL) for major defects shall be 1.5 percent and the AQL for minor defects shall be 2.5 percent. However, when major defects are considered collectively on a class basis, the AQL for any individual defect shall be limited to .40 percent. In addition, where three or less defects appear in a classification, sampling plans applicable to a class shall not be used for major defects. (see 6.6).

c. Critical defects. Unless otherwise specified, one hundred percent examination shall be performed for all critical defects. If a visual critical defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing), the lot shall be rejected. The lot shall then be screened and resubmitted for visual inspection of critical defects. If a critical defect is found during packing, the portion of the lot that has been packed or is in the process of being packed shall be rejected. In addition the portion of the lot remaining to be packed shall be rejected. The lot shall then be screened and resubmitted for visual inspection of critical defects.

d. MIL-STD-651 shall apply in defining and evaluating visual defects.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7258816
4.4.2.1	Rotating Band Blank				NEXT HIGHER ASSEMBLY 12576579
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Foreign matter		.25	3.11.2	Visual
102.	Defective metal, including lamination		.25	3.2	Visual
103.	Incomplete or deformed		.25	3.2	Visual 1/
Minor					
201.	.190 Width, max.		.40	3.2	11075094
202.	Poor workmanship		.65	3.11	Visual

NOTES: 1/ A rotating band blank which is deformed or is judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions, shall be classed defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12756581
4.4.2.2	Windshield				NEXT HIGHER ASSEMBLY 12576578
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	.778 Diameter, ogive		.25	3.2	Gage
102.	.750 Diameter, mouth I.D.		.25	3.2	Gage
103.	.030 Thickness wall		.25	3.2	Gage
Minor					
201.	.034 Thickness, ogive wall, min		.40	3.2	Gage
202.	Profile ogive		.40	3.2	Comparator
203.	Mutilated		.40	3.2	Visual
204.	Foreign matter		.40	3.11.2	Visual
205.	Poor workmanship		.65	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

MIL-P-70637(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 12576580
4.4.2.3	Shot, Body				NEXT HIGHER ASSEMBLY 12576579
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Rotating band seat knurling missing or incomplete		.25	3.2	Visual
102.	Windshield seat grooves or knurling missing or incomplete		.25	3.2	Visual
103.	.400 Length, base to rear of rotating band seat		.25	3.2	Gage
104.	.012 Depth, rotating band seat, knurling, min.		.25	3.2	Visual/Gage 1/
105.	.743 Diameter, windshield seat after grooving, min.		.25	3.2	Gage
106.	.008 Runout of .743 groove diameter with datum surface		.25	3.2	Gage
107.	.295 Diameter, tracer cavity, min.		.25	3.2	7275149
108.	.600 Depth, tracer cavity		.25	3.2	7286995
109.	.008 Runout, diameter tracer cavity with datum surface		.25	3.2	7558875/7558876

NOTES: 1/ Measurement of knurling for compliance with the minimum depth and width requirements shall apply where visual inspection indicates that the depth or width dimension is borderline.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70637(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 12576580
4.4.2.3	Shot, Body				NEXT HIGHER ASSEMBLY 12576579
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
110.	.770 Diameter, body forward of rotating band seat		.25	3.2	5055857
111.	Foreign matter in rotating band seat		.25	3.11.2	Visual
Minor					
201.	.295 Diameter, tracer cavity, max		.40	3.2	Gage
202.	.680 Diameter, rotating band seat knurling		.40	3.2	Gage 2/
203.	.198 Width, rotating band seat		.40	3.2	7259353/7259536
204.	.17 Width, rotating band seat knurling, min.		.40	3.2	7259467 1/
205.	Profile, ogive and groove location		.40	3.2	Comparator
206.	Chamfer, tracer cavity		.40	3.2	Visual
207.	Foreign matter		.40	3.11.2	Visual
208.	Poor Workmanship		.65	3.11	Visual

NOTES:

2/ Gage, Snap - (Anvils, .170 min. width)
(Go .680) .6796 - .0004
(Not go .662) .6620 + .0004

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70637(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 12576579
4.4.2.4	Body and Rotating Band Assembly				NEXT HIGHER ASSEMBLY 12576578
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Metal defect		100%	3.5	4.5.2/11075203 (REF)
Major					
101.	.008 Runout, body rear of rotating band with datum surface		.25	3.2	Gage
102.	.008 Runout, rotating band with datum surface		.25	3.2	7558875/7558876
103.	.828 Diameter, rotating band		.25	3.2	Gage 1/
104.	Profile, crimp groove		.25	3.2	Comparator/20x/8657201/ 7558294
105.	.005 Radius, rear edge of crimp groove, max		.25	3.2	Comparator/50x
106.	.195 Location, crimp groove		.25	3.2	Comparator/20x/8657201/ 7558294
107.	Void at junction of rotating band and projectile body		.25	3.4	Visual
108.	.060 Radius base of projectile		.25	3.2	Visual/Comparator/ 50x 2/
Minor					
201.	.766 Diameter, body rear of rotating band, min.		.40	3.2	5055857/7259718 Comparator
202.	Depth and width of undercut front and rear of rotating band		.40	3.2	7259718/Comparator/31.25X

NOTES: 1/ Ring gage - (Go .828) .82780 + .00012 Snap Gage - (Not go .824) .8240 + .0002

2/ Measurement of base radius, and surface roughness shall apply where visual inspection indicates that the maximum dimension is borderline.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70637(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 12576579
4.4.2.4	Body and Rotating Band Assembly				NEXT HIGHER ASSEMBLY 12576578
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
203.	.198 width, rotating band		.40	3.2	Comparator/20x/8657201 7558294
204.	Surface roughness		.40	3.2	Visual 2/
205.	Corrosion		.40	3.2	Visual
206.	Mutilated		.40	3.2	Visual
207.	Poor Workmanship		.65	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70637(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12576578
4.4.2.5	Body and Windshield Assembly				NEXT HIGHER ASSEMBLY 12576577
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101.	.784 Diameter, bourrelet, min.		.25	3.2	Gage 1/
102.	2.850 Length, windshield to rear edge of rotating band, max.		.25	3.2	7287005
103.	3.240 Length, overall, max.		.25	3.2	Gage
104.	.008 runout, bourrelet diameter with datum surface		.25	3.2	7558875/7558876
105.	.020 runout windshield with datum surface		.25	3.2	7558875/7558876
106.	1974 grains, weight		.25	3.2	Balance
107.	Foreign matter (prior to assembly)		.25	3.9.1	Visual
Minor					
201.	.784 diameter, bourrelet, max.		.40	3.2	Gage
202.	Loose windshield		.40	3.2	Manual 2/
203.	Mutilated		.40	3.2	Visual
204.	Corrosion		.40	3.2	Visual
205	Foreign matter		.40	3.11.2	Visual
206.	Poor Workmanship		.65	3.11	Visual

NOTES:
1/ Snap gage - (Not go .782) .7820 + .0001
2/ If the windshield moves with respect to the shot body under thumb and finger pressure applied in the lateral or axial direction, the assembly shall be classified defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70637(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Projectile, 20mm, APT, M95 M/P (painted)				12576577
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101.	.786 Diameter, bourrelet, max.		.25	3.2	8655592
102.	770 Diameter, rear of rotating band seat, max.		.25	3.2	5055857
Minor					
201.	Paint not dry		.40	3.2	Tactile
202.	Paint inadequate		.40	3.2	Visual
203.	Paint on rotating band		.40	3.2	Visual
204.	Marking incorrect, incomplete, illegible or missing		.40	3.2	Visual
205.	Foreign matter (tracer cavity and exterior)		.40	3.11.2	Visual
206.	Rotating band damaged		.40	3.2	Visual
207.	Poor Workmanship		.65	3.11	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Tests. The tests listed in Table II shall be conducted on each lot in compliance with the methods of inspection prescribed in 4.5. The samples shall be selected in such a manner as to be representative of the entire lot. The sample size and acceptance criteria for each test shall be as specified.

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TABLE II. PROJECTILE TESTS

CLASSIFICATION OF DEFECTS & TESTS MIL-P-70637(AR)

PARAGRAPH	TITLE: Projectile, 20mm, APT, M95		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Band tightness (Body & Rotating Band Assy)	315	1/	3.4	4.5.1
	Soundness verification (Body & Windshield Assy)	315	2/	3.5	4.5.4
	Automatic inspection effectiveness (PCS No. 1)		3/	3.5	4.5.3
	Phosphate coating (Proj. M/P before painting or standard panels)	3	4/	3.6.1	4.5.5
	Paint adhesion (Proj. M/P Assy painted)	13	5/	3.6.2	4.5.6
	Corrosion resistance (Proj. M/P Assy painted)	32	6/	3.6.3	4.5.7
	Metal parts security (Proj. M/P Assy painted)	250	7/	3.7	4.5.8
	Hardness (Shot Body)	13	8/	3.8	4.5.9
	Windshield torque (Body & Windshield Assy)	20	9/	3.9.3	4.5.10

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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1/ If two or more units of the sample have an average radial void between the seated band and the band seat, greater than 0.004 inch, the lot shall be rejected. If one unit of the sample has an average radial void, greater than 0.004 inch, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units have an average radial void greater than 0.004 inch, the lot shall be rejected.

2/ Rejection of one or more units of the sample by the flaw detector in the test for body metal soundness shall be cause for rejection of the lot.

3/ Failure of the flaw detector to reject PCS No. 1 shall be cause for rejection of all projectile metal parts inspected for "soundness" since the preceding test and shall require requalification of the equipment for automatic inspection. (see 4.5.3).

4/ Failure of the phosphate coating on the sample projectiles or standard panels to comply with the applicable requirements shall cause rejection of all production projectile metal parts phosphated since the preceding test.

5/ If two or more units of the sample fail to comply with the requirement for paint adhesion, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the requirement for paint adhesion, the lot shall be rejected.

6/ If two or more units of the sample fail to comply with the corrosion resistance requirement, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

7/ The lot shall be rejected, if any one of the following types of metal parts separation or breakup occurs in the metal parts security test, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s):

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a. Body shot breakup.

b. Separation of windshield from the shot body.

c. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint.

d. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint.

e. The following acceptance criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint:

(1) The lot shall be accepted if two or less projectile metal parts throw band fragments having length not greater than 0.500 inch.

(2) The lot shall be rejected if three or more projectile metal parts throw band fragments having length not greater than 0.500 inch.

(3) The lot shall be rejected if any projectile metal part throws a band fragment having length greater than 0.500 inch (indicative of deep-seated band lamination).

(4) The lot shall not be penalized for band particles thrown as a result of normal band fringing or slivering.

8/ If two or more units of the sample fail to comply with the body hardness requirements, the lot shall be rejected. If one unit fails to comply with the requirements, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the hardness requirements, the lot shall be rejected.

9/ Failure of two or more of the sample projectiles to comply with the specified torque requirement shall cause rejection of the lot. If one projectile of the sample fails to comply, a second sample of 20 projectiles shall be tested. The lot shall be rejected if two or more projectiles of the combined samples fail to comply with the applicable requirement.

4.4.4 Packaging, packing and marking inspection. Inspection for packaging, packing and marking shall comply with Dwg. 7258861-5 or Dwg. 7259136-5 and MIL-STD-1169.

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4.4.5 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Band tightness. The method of test shall comply with all requirements of the operating instructions for the band tightness tester Dwg. 7259545. The inspection sample shall be cycled through the tester to determine compliance with the rotating band tightness requirement. The total measurement of radial void registered on the tester indicator at final ram pressure, shall be divided by four (number of indentors) to determine the average radial void.

4.5.2 Soundness. The method of inspection shall comply with all requirements of Eddy Current Flaw Detector, Dwg. 11075203, (ref) except for those inspection requirements which are reserved for Government application, as specified herein.

4.5.2.1 Application of defect standards. Master defect standards, referenced in the Operating Instructions, shall apply to resolve inspection discrepancies occurring between the results of defect standards for contractor "soundness" inspection and those for Government "soundness" verification. (see 6.8).

4.5.3 Automatic eddy current inspection effectiveness. Equipment checkout shall be performed as stated in the operation instructions at the start of each shift and after any adjustment or shutdown. Upon completion of checkout, cycle the PCS No. 1. front slot and rear slot standard through the inspection system during automatic operation. All three standards must be rejected to assure correct adjustment for reject level and scan limits. During remainder of shift the PCS No. 1 shall be cycled through the equipment at intervals of 20 $\pm$ 5 minutes during automatic operation.

4.5.3.1 Equipment verification. The Government inspector shall maintain independent of the contractor, a set of standards and may cycle these standards through the equipment at any time. Failure of equipment to reject standards will be evidence that the equipment is not properly adjusted and the contractor will be so notified.

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4.5.4 Soundness verification. After equipment checkout and verification as specified in 4.5.3 and 4.5.3.1 the entire lot sample shall be inspected with the equipment operating automatically. Any rejected unit of the sample will be automatically cycled through the equipment with CAL. button pressed (to lower the sensitivity one step below that used for production inspection). Any unit which is rejected at the lower sensitivity shall be classed as a defective.

4.5.5 Phosphate coating. The method of test shall be as prescribed in TT-C-490 for the type of coating used (see Dwg. 12576577). The prescribed sample projectiles or standard test panels shall be processed at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.5.6 Paint adhesion. The method of test shall be as prescribed in TT-C-490. The test shall be made on the exterior painted surface of each projectile metal part of the sample forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal on any unit of the sample, as a result of the test, shall be recorded as a defect.

4.5.7 Corrosion resistance. The method of test shall be as prescribed in American Society for Testing and Materials (ASTM), ASTM B117. The paint shall not be disturbed, however, the rotating band and unpainted surfaces of the sample projectile metal part may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.6.3.

4.5.8 Metal parts security. The method of test will be as prescribed in AMCR 715-505, Volume 8, as applicable. The sample projectile metal parts will be loaded and assembled as specified on Dwg. 7259190-1. The test cartridges, at ambient temperature, will be fired in bursts of 50 shots in a M3 or MK16 gun. The gun barrel shall be at ambient temperature at the beginning of the test and cooled to ambient temperature after each 100 rounds fired. The cyclic rate of fire shall be recorded for informational purposes.

4.5.9 Heat treatment. No more than one composition of steel shall be used for the shot body in any one projectile lot. Not more than six heats of steel shall be used in the shot bodies in any one projectile lot, except that in the make-up of miscellaneous lots only the restriction on composition shall apply.

4.5.9.1 Hardness. The method of test shall be in accordance with ASTM E18, Rockwell superficial hardness of metallic materials.

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4.5.10 Windshield torque. The body and windshield assembly shall be marked with a light scratch extending axially across body onto the windshield. Then the asssembly shall be placed in a suitable fixture and subject the contacting areas of the body and windshield to 1000 inch pound torque applied slowly to the projectile. Observation shall be made for movement of the windshield with respect to the misalignment in the scratch mark. The nose end of the windshield of the test projectiles may be modified to facilitate application of test loads.

4.5.11 Penetration. The V50 protection ballistic limit test shall be performed in accordance with TECOM Top 2-2-710.

5. PACKAGING

5.1 Preservation and packaging. The projectiles shall be preserved and packaged in accordance with Dwg. 7258861-5 or Dwg. 7259136-5, as directed by the procuring activity.

5.2 Packing. The projectiles shall be preserved and packaged as specified in 5.1. Each lot shall be packed for shipment in accordance with Dwg. 7258861-5 or Dwg. 7259136-5, as directed by the procuring activity.

5.3 Marking. Marking or labeling of packed cartons shall comply with applicable requirements of MIL-STD-1169.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use with Cartridge, 20MM, AP-T, M95

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDEC ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements Lists, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.5 First article sample. Where a first article sample is required by the contract, the contracting officer will notify the test facility when to proceed with the metal parts security test. Additional assemblies may be required by the test facility. (see 4.3)

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6.6. Intermediate point inspection. The classification of defects identifies the defect characteristics (among other things) for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which would be detrimental to the attainment of optimum product quality of the end item.

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, US Army Armament Research and Development Command (ARRADCOM) or US Army Armament Research and Developement Center (ARDC) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.8 Defect standard. The contractor will be responsible for the procurement, manufacture, and maintenance of the eddy current equipment and the front slot, rear slot, phase set, and three slot standards IAW the latest drawings called out in the applicable technical data package and the contract. The production check E.C. Standards (PCS No. 1) Dwg. 11075245 will be furnished by the contractor to the Government agency having contract administration responsibility.

6.9 Subject term (key word) listing.

Armor Piercing Tracer
Mk16 20mm Automatic Weapon System
Small Arms Ammunition

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-AC71)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70637	Projectile, 20MM, Armor Piercing-Tracer, M95, Metal Parts

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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